

National Systems of Innovation

Toward a Theory of Innovation
and Interactive Learning

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Toward a Theory of Innovation and Interactive Learning

Edited by
Bengt-Åke Lundvall



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PREFACE

This book has a long and complex history. Already in the late seventies, the IKE-group at Aalborg University began to integrate a French structuralist approach to national systems of production with the Anglosaxon tradition in innovation studies, in order to explain international competitiveness. This 'new combination' is reflected in the concept national systems of innovation which is at the centre of the book. The first sketches to the book, which date back to 1986, were specifically oriented towards the problems of small national systems of innovation (Andersen and Lundvall, 1988), but gradually we realised that the process of internationalisation and globalisation has made all countries 'small', and this is reflected in the final design of the book.

At several stages of the project, the Danish Social Science Research Council has given its financial support. The Research Committee, Aalborg University has funded some of the activities while our department, Institute for Production, has been consistently supportive both financially and administratively. We gratefully acknowledge the patience of these funding institutions.

During the long process, many people have commented on the general outline and on drafts of chapters at workshops and seminars. In particular, we would like to thank Carlota Perez for her often very critical but still supportive comments, at the early stage of the project, and Cristopher Freeman who is not only a co-author, but also has given us invaluable advice all through the process. We are indebted also to Gerhard Bräunling, Charles Edquist, Anders Joest Hingel, Poul Thojs Madsen, Lars Mjøset, Irene Odgaard, Jørgen Lindgaard Pedersen, Keith Smith, Lennart Stenberg and Kjell Trykkestad who, at a workshop in May 1991, commented the drafts for the book. A special thanks to Maureen O'Kelvey and Mark Elam who have tried to polish on our primitive English and who luckily could not restrain themselves from giving useful comments also to the substance.

Dorte Køster, the secretary of the IKE-group, has as always been able to solve difficult coordinative tasks in a simple and efficient way. We are grateful to her and to MA-students Keld Laursen and Ina Drejer who, in the final phase, helped a lot with the detailed editing of the chapters.

The editing process has as most activities in the IKE-group been a genuinely interactive and collective process. Björn Johnson has co-edited part I, Esben Sloth Andersen part II and Bent Dalum part III. Without their generous help the publication of this book might have had to await the turn of the century.

The editor

Aalborg, 25 February 1992

Chapter 1

INTRODUCTION

Bengt-Åke Lundvall

1.1. Introduction

Theories in the social sciences may be regarded as ‘focusing devices’. Any specific theory brings forward and exposes some aspects of the real world, leaving others in obscurity. That is why a long lasting hegemony of one single theoretical tradition is damaging both in terms of understanding and policy-making. In the field of economics, the dominating neo-classical paradigm puts its analytical focus upon concepts such as scarcity, allocation, and exchange, in a static context. Even if these concepts reflect important phenomena in the real world, they only bring forward some aspects of the economic system. One aim of this book is to demonstrate the need for an alternative, and supplementary, focusing device which puts interactive learning and innovation at the centre of analysis.

Through more than a decade, a group of economists at Aalborg University, the IKE-group, has worked together studying industrial development and international competitiveness from such a perspective. This book presents results from this work in relation to one specific subject; national systems of innovation.¹

Our choice of perspective and subject is based upon two sets of assumptions.

First, it is assumed that the most fundamental resource in the modern economy is knowledge and, accordingly, that the most important process is learning. The fact that knowledge differs in crucial respects from other resources in the economy makes standard economics less relevant and motivates efforts to develop an alternative paradigm.²

Second, it is assumed that learning is predominantly an interactive and, therefore, a socially embedded process which cannot be understood without taking into consideration its institutional and cultural context. Specifically, it is assumed that the historical establishment and development of the modern nation state was a necessary prerequisite for the acceleration of the process of

learning which propelled the process of industrialisation, in the last centuries. Finally, it is recognised that the traditional role of nation states in supporting learning processes is now challenged by the process of internationalisation and globalisation.

These ideas are reflected in the overall structure of the book which is divided into three main parts. The first part presents the theoretical framework, the second part analyses the most important elements of the system of innovation and the third part is devoted to the opening of national systems through internationalisation and globalisation. This introductory chapter presents basic definitions, theoretical starting points, a road map for the book as a whole and, finally, references to other attempts to analyse national systems of innovation.

1.2. National Systems of Innovation

1.2.1. A First Definition

According to Boulding (1985), the broadest possible definition of a system is ‘anything that is not chaos’. Somewhat more specifically, a system is constituted by a number of elements and by the relationships between these elements. It follows that a system of innovation is constituted by elements and relationships which interact in the production, diffusion and use of new, and economically useful, knowledge and that a national system encompasses elements and relationships, either located within or rooted inside the borders of a nation state.³

Using the terminology of Boulding, it is obvious that the national system of innovation is a *social* system. A central activity in the system of innovation is learning, and learning is a social activity, which involves interaction between people. It is also a *dynamic* system, characterised both by positive feed-back and by reproduction. Often, the elements of the system of innovation either reinforce each other in promoting processes of learning and innovation or, conversely, combine into constellations blocking such processes. Cumulative causation, and virtuous and vicious circles, are characteristics of systems and sub-systems of innovation. Another important aspect of the innovation system relates to the reproduction of the knowledge of individuals or collective agents (through remembering).

1.2.2. Nation States and National Systems

The concept, national systems of innovation, presumes the existence of nation states and this phenomenon has two dimensions; the national-cultural and the étatist-political. The ideal, abstract, nation state is one where the two dimensions coincide, i.e. where all individuals belonging to a nation – defined by cultural,

ethnic and linguistic characteristics – are gathered in one single geographical space controlled by one central state authority (without foreign nationalities).

It is difficult to find any nation states, in this strict sense, in the real world. Countries differ both in the degree of cultural homogeneity and in the degree of political centralisation. In some cases it is not even clear where to locate the borders of a ‘national’ system of innovation. This might be true both for ‘multinational’ states as Belgium, Canada and Switzerland and for single-national but federal states such as Germany. At the extreme, a country might be solely constituted by a joint foreign policy with little in common in terms of its institutional set up and culture. In such cases, the concept of a ‘national’ system of innovation would be of little relevance.

Most of the contributors to this book have their roots in a minority of small countries which may be characterised as culturally homogeneous and socio-economically coherent systems (Sweden, Denmark and Norway). This gives a certain bias to our world outlook (it should do so, according to our basic understanding where theoretical conceptualisation is assumed to be culturally bounded). On the other hand, it may be argued, it is quite useful, analytically, to use concepts which are archetypes rather than ‘averages’. In order to bring out sharply the limits and consequences of globalisation and regionalisation, it is useful, at least as a starting point, to assume countries to be homogeneous in political and cultural terms.

1.2.3. National Systems, Globalisation and Regionalisation

Readers might ask, why we focus on the *national* level, in an era where many analysts point to an accelerating process of internationalisation and globalisation, characterised by multinational firms, loosening their relations to their home-country and entering into alliances with foreign firms. This process might actually be most advanced when it comes to the production of new knowledge and innovations in science-based technologies such as bio-technology, pharmaceuticals and electronics.

At the same time, a growing number of social scientists – often inspired by new sets of ideas labeled ‘flexible specialisation’, ‘networking’ and ‘post-fordism’ – have argued that regional production systems, industrial districts and technological districts are becoming increasingly important. Some authors analyse these two tendencies as interconnected and mutually reinforcing (Storper, 1991a, Camagni, 1990 and also Porter, 1990). They assume that globalisation, and international specialisation have their roots in the strengthening of specialised technological districts and regional networks.

Both globalisation and regionalisation, might be interpreted as processes which weaken the coherence and importance of national systems. In this

book, we do not deny the validity of these trends. Actually, we think that they make it even more pertinent to understand the role and workings of national systems of innovation, both historically, and in the present era.

First, we believe that national systems still play an important role in supporting and directing processes of innovation and learning. The uncertainties involved in innovation and the importance of learning imply that the process calls for a complex communication between the parties involved. This will especially be the case when the knowledge exchanged is tacit and difficult to codify. When the parties involved originate in the same national environment – sharing its norms and culturally based system of interpretation – interactive learning and innovation will be easier to develop.

On the other hand, it must be recognised that important elements of the process of innovation tend to become transnational and global rather than national – and here the trend will be most important in science-based areas where the communication is easier to formalise and codify. Some of the big corporations are weakening their ties to their home-base country and begin to spread their innovative activities and to ‘source’ different national systems of innovation. These changes are important and they challenge the traditional role of national systems of innovation, but they do not make it less important to understand how national systems work.

When an old institutional order is threatened, and a new one is beginning to develop, it becomes critically important to understand the basic mechanisms of the old order. Without such an understanding the costs of transformation might become unnecessarily high. More specifically, the process of far-reaching European integration may run into serious problems if it does not take into account the complex interaction between institutions and economic structure in promoting innovation at the national level.

Behind the analysis lies also, as mentioned, the hypothesis that the modern nation states in the Western world – not necessarily the new states in the former colonies – have worked as ‘engines of growth’. They were constituted and shaped in their present form in a period characterised by a rapid economic transformation, including the massive movement of labour from agriculture to industrial production. Their social institutions and state policies have supported such a transformation and new institutions aiming directly at economic wealth creation through innovation have been established in the course of the last century and become integral parts of national systems of production.

From what has been said, it is obvious that national systems of innovation are open and heterogeneous systems. Processes of innovation transcend national borders and sometimes they are local rather than national. Actually, this has always been the case for most national systems. The rapid industrialisation and modernisation of European countries, starting more than 100 years ago, was

closely connected with an opening up of the national economies in terms of foreign trade, capital import and import of foreign ideas and experts, and already at that time the international specialisation was often reflected in a regional specialisation within the countries.

1.2.4. Public Policy and National Systems of Innovation

As pointed out at the very beginning of this introduction, one main purpose of this book is to contribute to a theoretical understanding of interactive learning and innovation. But the concept 'national systems of innovation' may also be useful when it comes to inspire public policies at the national and the international level.

First, in order to determine what governments should do in order to promote innovation, it is useful to know the specific systemic context in which a national government intervenes. Otherwise, government policies might either reproduce weaknesses of the national system or introduce mechanisms incompatible with the basic logic of the system.

Second, in the increasingly serious international conflicts about which countries are paying for (the US) and, respectively, appropriating benefits from (Japan) the investment in science and development of new technology, it is important to understand how different and very diverse national systems work. This is a point made by experts close to the GATT-negotiations (Ostry, 1990).

Third, in a world characterised by a radical shift in techno-economic foundations, the ability of national systems to cope, successfully, with change and to exploit new technical opportunities seem to be quite divergent (Freeman and Perez, 1988). Learning from the experience of foreign systems, in this respect, might be facilitated if the workings of the respective national systems as a whole are properly understood. Strategies based on naive copying may be avoided and institutional learning across national borders might be stimulated (See also chapter 14). Not least, the present development in Eastern Europe points to a strong need to develop a realistic understanding of the workings of the 'real market economies' in relation to innovation.

Actually, the concept, national systems of innovation, has already entered the vocabulary of policy-makers at the national and the international level. An ambitious effort to understand the importance of technology for economic change, was launched by OECD in 1988, as the Technology/Economy programme (TEP). When the outcome of this programme was summed up in Montreal, in 1991, the concept, national systems of innovation, was given a prominent place in the conclusions.

It was pointed out that the assignment of proper roles for, respectively, government and the private sector in enhancing technological capabilities should

build upon a better understanding of national systems of innovation. Also, it was concluded that the growing international conflicts regarding the global sharing of burdens and benefits emanating from the development and use of new technology might be kept within reasonable limits only if the parties get a better understanding of the diversity of NSI's (OECD – Canada, 1991). The fact that the concept has already entered the everyday vocabulary of policy-makers makes it even more important to give the NSI-concept an analytical basis.

1.2.5. Performance of National Systems of Innovation

In order to design policies relevant for national system of innovation, it is necessary to agree upon which should be the 'desiderata' of the system (Kornai, 1971, 214 f). From the standpoint of general equilibrium theory, the main performance dimension refers to the more or less efficient allocation of scarce and given resources. A more dynamic version would point to the adaptability of the system. A Keynesian perspective would emphasise the degree of utilisation of existing resources and especially of the labour force.

At this general level, we would like to propose that the most relevant performance indicators of national system of innovation should reflect the efficiency and effectiveness in producing, diffusing and exploiting economically useful knowledge. Such indicators are not well developed today. One of the classical measures for comparing different national systems is R&D-expenditure as a proportion of GDP. There are two obvious problems with this indicator. First, it reflects only an input effort and does not say anything about what comes out of the effort. Second, R&D-expenditure is only one kind of relevant input to the process of innovation – learning in connection with routine activities may be more important than R&D.

The output measures used are more recently developed and include patents (Pavitt and Patel, 1988), the proportion of new products in sales (Kristensen and Lundvall, 1991) and the proportion of hi-tech products in foreign trade (Dalum et al., 1988). Each one of these indicators has its own specific weaknesses and it is wise to combine them in order to get a more satisfactory picture of the performance of a national system. A common weakness is that these measures do not take into account the diffusion of process technology and in order to get a more complete picture, indicators for diffusion should be taken into account (Edquist and Jakobsson, 1988).

Technical progress is not regarded as a goal in itself. The main reason why national governments engage in innovation policy is the assumption that innovation is a key element in national economic growth. Different indicators of economic growth (national income or consumption per capita) are relevant when it comes to compare systems. But such indicators will reflect factors which have

little to do with innovation and, more importantly, they give little insight into how innovation takes place in different countries. One interesting observation is that different systems may develop different modes of innovation while still following parallel growth paths.⁴ Bringing the different, specific innovation indicators into the picture helps to characterise the specific national mode of innovation.

1.2.6. The Normative Dimension

The choice of performance criteria and of the respective weights to be assigned to them is fundamentally normative decisions. One of the most developed attempts to come to grips with implicit and explicit value judgements in economic analysis is Myrdal (1968). He argues that as a minimum requirement economists should make explicit their value premises. Further, Myrdal actually presents a method for bringing value premises into the analysis of national economies. When studying the problems of the poor Asian countries, he chooses to accept the set of value premises predominating among the national establishment in the countries studied – the ideal of modernisation.

Given the lack of alternatives, it is tempting to use a similar approach to national systems of innovation. To identify the ambitions and goals of national governments in the area of innovation is, apparently, quite easy. The public discourse is dominated by references to the international competitiveness of the national economy and to national economic growth. There seems to be a broad social acceptance in the national establishment that these are the main goals. But this is not the only relevant level to take into account.

Another level of analysis refers to the international organisations of the rich countries such as the European Community and OECD. Politicians and experts at this level are more oriented towards strengthening economic growth in their respective region and towards avoiding international conflicts within the community of countries they represent.

Finally, there is a global level of analysis, with a rather weak representation in organisational terms – the UN organisations, global environmentalist organisations etc. At this level, it becomes more obvious to experts and politicians that the long term survival of the global economy is dependent upon ecological sustainability and upon a reduction of the extreme social inequality at level.

We do not find it proper to adopt the set of value premises of any single one of these three levels, however. On the one hand, we consider national policies and goals, relating to innovativeness and competitiveness to be legitimate, to a certain degree. The pursuit of such goals has been an important motor behind the dramatic increase in economic wealth in the OECD-area and in some newly industrialised countries in Asia. Additionally, policies designed to

strengthen the system of innovation are less of ‘beggar-thy-neighbor’ character than exchange rate or incomes policies.

On the other hand, we realise that some of the games related to national science and technology policy may actually be zero-sum games and that there is a growing number of examples of unpleasant trade-offs between short-term national economic growth and long term global sustainability (in terms of environment, natural resources, etc.). The national context tends to become too narrow when it comes to solving problems such as global inequity and sustainability. The value premises of the national establishment must be confronted with these broader and more long term concerns.

This is one reason, why we welcome the development and strengthening of organisations and agencies operating at the international and global level. But, as already pointed out, it might be premature to dismantle the nation states and the national systems of innovation. The on-going process of innovation changes the social conditions of citizens and regions – some for the better and some for the worse. The most important role of the nation state in this context has been to compensate the weak and to put some restraint on the strong. Without the formation of new agencies, capable of realising such a ‘social dimension’, a process of internationalisation and globalisation, which undermines national systems of innovation, might result in a long term social and political crisis rather than in creative destruction.

1.3. Towards a Theory

1.3.1. Innovation as a Cumulative Process

In the models of standard economics, innovations appear as extraordinary events, coming from the outside, which temporarily disturb the general equilibrium. After a process of adjustment, reflecting the work of the price mechanism, a new state of equilibrium is established. This approach might have been adequate in pre-industrial societies where innovations seemed to occur as rare and exogenous events. In modern capitalism, however, innovation is a fundamental and inherent phenomenon; the long-term competitiveness of firms, and of national economies, reflect their innovative capability and, moreover, firms must engage in activities which aim at innovation just in order to hold their ground.

One of our starting points is that innovation is a ubiquitous phenomenon in the modern economy. In practically all parts of the economy, and at all times, we expect to find on-going processes of learning, searching and exploring, which result in new products, new techniques, new forms of organisation and new

markets. In some parts of the economy, these activities might be slow, gradual and incremental, but they will still be there if we take a closer look.

The first step in recognising innovation as an ubiquitous phenomenon is to focus upon its gradual and cumulative aspects. Such a perspective gives rise to simple hypotheses about the dependence of future innovation on the past. In this context an innovation may be regarded as a new use of pre-existing possibilities and components. Here, Schumpeter's choice of terminology, where 'innovations' and 'new combinations' are used as synonyms, is enlightening. Almost all innovations reflect already existing knowledge, combined in new ways.

This is, however, not to say that the step to the new combination is always of the same character. Sometimes, an innovation might be almost inevitable – the new combination might be easy to find and to realise. In other cases, it might take an enormous intellectual effort or an extremely creative mind, to identify a potential new combination. And, sometimes, the process of innovation results in radical breaks with the past, making a substantial part of accumulated knowledge obsolete. Another of Schumpeter's concepts, 'creative destruction', points to this discontinuity and it might be applied not only to the structure of production, but also to the structure of knowledge.

Nevertheless, we will put some emphasis upon the ubiquitous and cumulative character of innovation. In such a perspective the distinction made in innovation theory, between invention, innovation, and diffusion, as three separate stages necessarily becomes blurred. We also understand why it is difficult to date invention and innovation in time, and why an innovation does not stay the same throughout its diffusion. Innovation appears now, not primarily as a single event, but rather *as a process*.

A second starting point is that interactive learning and collective entrepreneurship are fundamental to the process of innovation. In his early work on the theory of economic development, Schumpeter pointed to entrepreneurs, which act individually, as the most important economic agents bringing innovations into the economic system (Schumpeter, 1934). Later, he revised his theoretical scheme, however, by giving a critical role to the collective work in R&D-laboratories (Schumpeter, 1942). In a sense, through introducing systems of innovation we pursue this trajectory from individual towards collective entrepreneurship further.

We will argue that most important forms of learning may fundamentally be regarded as interactive processes, and that together the economic structure and the institutional set up form the framework for, and strongly affect, processes of interactive learning, sometimes resulting in innovations. In the chapters to follow this general point will be illustrated at different levels of analysis.

1.3.2. Learning and the Structure of Production

One of the most important institutional innovations in the last century was the establishment of R&D-laboratories in the big private firms (Freeman, 1982, and chapter 9 in this book). Scientific activities and technical change have been brought closer together and become increasingly interdependent activities and today, the capability to innovate cannot be assessed in isolation from efforts in science, research and development. However, here, we will insist upon the fact that not all important inputs to the process of innovation emanate from science and R&D-efforts. We, thus, assume that learning takes place in connection with routine activities in production, distribution and consumption, and produces important inputs to the process of innovation. The everyday-experiences of workers, production engineers, and sales representatives influence the *agenda* determining the direction of innovative efforts, and they *produce knowledge and insights* forming crucial inputs to the process of innovation.

When bottle-neck problems are met and registered in production, or in the use of a product, the agendas of producers change, affecting the direction of their innovation efforts. Everyday-experience also increases technical knowledge and gives ideas about in which direction solutions should be looked for. Such activities involve learning-by-doing, increasing the efficiency of production operations (Arrow 1962), learning-by-using increasing the efficiency of the use of complex systems (Rosenberg, 1982), and learning-by-interacting, involving users and producers in an interaction resulting in product innovations (Lundvall, 1988).

If innovation reflects learning, and if learning partially emanates from routine activities, innovation must be *rooted in the prevailing economic structure*. The areas where technical advance will take place, will primarily be those where a firm, or a national economy, is already engaged in routine activities.⁵ This basic idea is developed in *chapter 4* where the focus is upon how the national system of innovation is rooted in a national system of production.

1.3.3. Learning and the Institutional Set Up

The institutional set-up (of a specific firm, a constellation of firms, or a nation) is the second important dimension of the system of innovation. Institutions provide agents and collectives with guide-posts for action. In a world characterised by innovative activities, uncertainty will be an important aspect of economic life. Institutions make it possible for economic systems to survive and act in an uncertain world. Institutions may be routines, guiding everyday actions in production, distribution and consumption, but they may also be guide-posts for change. In this context, we may regard technological trajectories and paradigms, which focus the innovative activities of scientists, engineers, and technicians, as one special kind of institutions.⁶

One of the fundamental characteristics of institutions is their relative stability over time. They arise because, in a changing and uncertain world, agents and organisations need guidance and institutions make life more manageable and comfortable (not necessarily more efficient in any sense of this term) for them. This perspective is discussed in *chapter 2* where it is demonstrated that institutions and routines are fundamental, because they provide the stability needed for innovative efforts to take place and to be successful.

1.3.4. Product Innovation and User-Producer Interaction

One way to illustrate how the structure of production and the institutional set up, together, affect the rate and direction of innovation is to focus upon product innovations, and their roots in the interaction between producers and users. First, at the micro level, the structure of production defines sets of user-producer relationships, which condition the scope and direction of the process of innovation. Second, the institutional form which characterises these relationships – and, especially, the elements of organisation in these markets – reflects the characteristics of the process of innovation. Third, the institutional set up, once established, will affect the rate and direction of innovation. Fourth, one interesting dimension of user-producer relationships can be shown to be distance in cultural and geographical space.

Through this analysis, developed in *chapter 3*, we illustrate how the production structure, and the institutional set up jointly define a system of innovation and at the same time provide an understanding of the micro-foundation of ‘national systems of innovation’.

1.3.5. Learning, Searching and Exploring⁷

Above, we have indicated the importance of learning rooted in routine activities. But, of course, economic agents and organisations also consciously invest time and resources in expanding their technical knowledge. *Searching* is another important activity, creating inputs to the system of innovation.

Organisations normally learning only from routine activities of production and distribution might engage in search activities under certain extreme circumstances. When the survival of the organisation is threatened, its members become engaged in what might be called ‘desperate search’. This kind of search might begin as local search, looking for alternatives (in terms of products, processes, markets etc.) close to the ones already well-known to the organisation and, only if it is impossible to find any satisfactory solutions in this area, expanding to more distant alternatives.⁸

However, desperate search is not always very efficient. Especially when technology is science-based, complex, and changing, it becomes attractive to establish special departments permanently engaged in searching activities. Organisations with searching as their special mission might be departments for market analysis as well as R&D-departments, and – laboratories.

Searching which takes place in academic or science-oriented organisations, outside the private firms, brings forward another kind of raw material for the process of innovation. We call this kind of search ‘exploring’. The most important difference between exploring and searching is that ‘exploring’ is less goal-oriented than profit oriented search. If we take a closer look at scientific activities we shall, however, often find that they too have a specific aim and direction. Even if the paradigms and trajectories, which determine the aims and directions of basic science, develop more according to their own internal logic and are less responsive to changes in economic parameters than innovations in private firms, the producers of basic science will, to a certain extent, be oriented towards users outside the realm of pure science. The direction of research in mathematics and logic may, for example, reflect new needs developed by computer scientists and soft-ware experts.

Exploring will, because of its weaker goal-orientation, sometimes result in outcomes, neither foreseen, nor looked for, by profit oriented organisations. This adds to technological change a dimension of dynamism and radical change, extremely important in the long run. Exploring will sometimes result in breaks in cumulative paths and create the basis for new technological paradigms.

1.3.6. Incremental Versus Radical Innovations

If innovation is rooted in learning, and learning in routine activities, we might expect all innovative activities to be incremental, and it would be rather simple to predict the direction of technical change. But we must take into account the fundamentally uncertain and disruptive character of the process of innovation. As mentioned, uncertainty rules in the process of scientific activities, and these produce, from time to time, results which were neither anticipated nor looked for. Uncertainty also rules regarding the economic impact of an innovation. A new product might fail either for technical reasons, or because it does not successfully address potential user needs. Conversely, a product originally addressed towards the needs of a small sub-set of users might later prove to be a commercial success, applicable in very substantial parts of the economy.

When distinguishing between incremental and radical innovations, we may refer, primarily, either to the technical or to the economic dimension. Some innovations, incremental in technical terms, may have a crucial impact upon the economy. This will be true for a small technical change solving a bottle-neck

problem of strategic importance (the introduction of vehicles with inflated rubber wheels in agriculture did not represent any radical technical break but it had a dramatic impact upon the productivity of this sector). On the other hand, an innovation very radical in technical terms, and signalling a new technological paradigm, might for technical reasons be premature and have a very limited impact on the economy (it took the Babbage-version of the computer, obviously a radical innovation in technical terms, more than a century before it had any economic impact at all). It follows that many radical innovations will be radical only in one of the two dimensions while remaining incremental in the other dimension.

For these reasons, we assume that the process of innovation is neither totally accidental nor totally predetermined by the economic structure and the institutional set up. The analysis of systems of innovation helps us to understand and explain, why technology develops in a certain direction, and at a certain rate, but a strong element of randomness will always remain.

1.3.7. Defining the NSI – The Role of Theory and History

From what has been said, it follows that we may make a distinction between a system of innovation in the narrow sense and a system of innovation in the broad sense. The narrow definition would include organisations and institutions involved in searching and exploring – such as R&D-departments, technological institutes and universities. The broad definition which follows from the theoretical perspective presented above includes all parts and aspects of the economic structure and the institutional set up affecting learning as well as searching and exploring – the production system, the marketing system and the system of finance present themselves as sub-systems in which learning takes place.

Determining in detail which sub-systems and social institutions should be included, or excluded, in the analysis of the system is a task involving historical analysis as well as theoretical considerations.

In different historical periods different parts of the economic system, or different inter-faces between sub-systems, may play a more or less important role in the process of innovation. In the early British industrialisation new technology reflected, primarily, learning inside firms which developed and tested new production equipment, either developed in-house, or in cooperation with artisans from small workshops. The development of the new industries of chemistry and electricity at the end of the last century changed the location of the innovation-nexus and brought it closer to the R&D-laboratories of big firms.

Today, it seems as if the crucial interfaces of systems of innovation have shifted again. Radical innovations in information technology which are, themselves, science-based have put the focus upon the coupling of routine based

learning to searching and R&D. ‘The factory as a laboratory’ – formula referred to in chapter 9 reflects these new trends. At the same time, however, more and more innovative activities have to draw upon quite different and separate sets of generic knowledge (bio-technology, microelectronics, new materials) making the process of science-based innovation, even more costly and complex.

However, the theoretical perspective is also important. The broad definition of the system used in this book reflects the importance attached to interactive learning as a basis for innovation. Alternatively a ‘linear model of technical change’ – where technical innovations were assumed to follow mechanically from scientific efforts and from research efforts inside firms – would define the system of innovation much more narrowly and identify it with the R&D-system.

From this background, it should be obvious that a definition of the system of innovation must, to a certain degree, be kept open and flexible regarding which sub-systems should be included and which processes should be studied. It also follows that we cannot insist upon one single approach to the national system of innovation as the only legitimate one. Different theoretical perspectives bring forward different aspects of the system.

1.4. The Elements of the System

In the real world the state and the public sector are rooted in national states, and their geographical sphere of influence is defined by national borders. The focus upon *national* systems reflects the fact that national economies differ regarding the structure of the production system and regarding the general institutional set up. Specifically, we assume that basic differences in historical experience, language, and culture will be reflected in national idiosyncrasies in:

- Internal organisation of firms
- Inter-firm relationships
- Role of the public sector
- Institutional set up of the financial sector
- R&D-intensity and R&D-organisation

International differences in these elements are important for the working of the system as a whole, but the relationships between the elements are just as important. For example both the organisation and strategies of the public sector, including its responsibility for education and R&D, and the financial sector will affect the way firms organise and form networks. On the other hand the historical specialisation of firms and networks of firms will be reflected in the public infrastructure of education and R&D.

First, we assume that the internal organisation of private firms is one important aspect of the system of innovation. Most innovations are developed by firms, and many innovation studies have demonstrated that the organisation of the flow of information, and of the learning process, are important and affect the innovative capability of the firm. The interaction between different departments engaged in respectively sales, production and R&D is one important aspect of the organisation which is attracting a growing interest in comparative innovation studies. Different aspects of the contingency of learning upon the organisation of work related to the innovation design dilemma are analysed in the chapter by Næs Gjerding (*chapter 5*).

But we also assume that inter-firm relationships are important in structuring the system of innovation. In standard economics these relationships are assumed to be characterised by competition and by pure markets. Focusing upon innovation makes it clear that cooperation between firms is a necessary supplement to competition, and one form of cooperation is the user-producer interaction analysed in chapter 4. In an increasing number of knowledge-intensive industries, other forms of inter-firm cooperation tend to become increasingly important. In the chapter by Gelsing (*chapter 6*), inter-firm interaction is presented in the context of network relationships and industrial districts, including informal exchange of technical know-how.

The public sector plays an important role in the process of innovation. As pointed out in chapter 9, it is involved in direct support of science and development, and its regulations and standards influence the rate and direction of innovation, and it is the single most important user of innovations developed in the private sector. In the chapter by Gregersen (*chapter 7*) the emphasis is put upon the role of the public sector as a more or less competent user of innovations.

The connection between the financial system and the process of innovation was strongly emphasised by the early Schumpeter and recently there has been a growing public interest in the enabling role of the financial system in regard to innovation. In the chapter by Lindgaard Christensen (*chapter 8*), the relationships between finance and innovation are analysed and interesting national differences are pointed out.

As already noted, the process of innovation is closely connected to the R&D-system; its resources, competencies and organisation. The historical development of the R&D-system and its current importance, is discussed in the chapter by Freeman (*chapter 9*). The chapter points to the need for institutional innovation in national systems of innovation.

Missing among these elements is the national education and training system. For different reasons, this extremely important element of the national system of innovation has not been given its proper treatment in

this book. There are big differences between countries in their formal and informal education and training systems which affect their innovative capabilities. This refers to the quantitative investment in education, the enrolment in science and engineering, the investment in training of skilled workers etc. Other differences are qualitative and also relate to the social norms and values reproduced by the system and to the degree of egalitarianism versus élitism in the system. An important task for future research is to integrate education and training systems and innovation systems in one single analytical framework.⁹

1.5. Opening the System

As pointed out, we do not assume the process of innovation to be exclusively localised inside national borders. On the contrary, we recognise that the process of innovation has increasingly become multinational and transnational reflecting, for example, R&D-cooperation between big firms based in different nations. Part III of the book analyses different aspects of the international relationships and takes into account that national systems of innovation are open and, increasingly, becoming wide-open.

In the chapter by Dalum (*chapter 10*), some important structural features of national systems of innovation are characterised by data on export specialisation. The long term development patterns of specialisation are analysed in terms of industry life cycles; it is shown that patterns are distinctly different country by country. The specialisation patterns are discussed in relation to structural competitiveness and in this context the strategic role of the engineering sector is emphasised.

Fagerberg (*chapter 11*) focuses upon one branch of trade theory with special relevance for the analysis of national systems of innovation, the home market theory. The relative importance of linkages between export specialisation in user- and producer industries are tested econometrically. This chapter and the one by Dalum are focused on trade patterns (arms' length trade) and do not take into account foreign direct investment and multinational enterprises.

Andersen and Brændgaard (*chapter 12*) present an analysis of economic integration from an evolutionary stand and study what might be regarded as an embryonic transnational system of innovation. The empirical case relates to the specialisation and competitiveness of the European Community in information technology and refers to some of the political efforts to strengthen the European Community in this field in relation to Japan and United States.

While these three chapters focus upon internationalisation in the form of international trade and specialisation, the chapter by Chesnais (*chapter 13*) brings foreign direct investment and multinational capital into the picture. The chapter

describes and analyses how the process of internationalisation has entered a new phase of globalisation, which fundamentally changes the role of national systems of innovation. Chesnais shows that the new tendencies challenge the relative autonomy of national systems and weaken their coherence, but also give a new and even more important role to public policy.

1.6. Alternative Approaches and Methods

1.6.1. Introduction

As pointed out, the definition of the national system of innovation is dependent upon the theoretical approach and therefore it is useful to see how different authors have used the concept and to relate the approach of this book to these alternatives.¹⁰

Friedrich List

The first systematic and theoretically based attempt to focus upon national systems of innovation goes back to Friedrich List (1841/1959). His contribution is also interesting because it is developed as an explicit alternative to Adam Smith and his contemporary followers. List makes a distinction between Adam Smith's 'cosmopolitan' approach which puts the focus upon exchange and allocation and his own national perspective focusing on the development of productive forces. We think that this is a fruitful and interesting distinction.

The only element of List's quite complex and rich – sometimes somewhat confusing – analysis still left in modern economics is his argument for protection of 'infant-industries'. His analysis went much further, however, indicating the need for governmental responsibility for education and training and for developing an infra-structure supporting industrial development. Actually, he sketched some of the most important elements of the national system of innovation.

Christopher Freeman

The first explicit use of the concept national systems of innovation may be the one in Freeman's (1987) book on Japan. Here the concept refers both to the nation-specific organisation of sub-systems and to the interaction between sub-systems. The organisation of R&D and of production in firms, the inter-firm relationships and the role of government and MITI are at the centre of the analysis which is both historical and based upon modern innovation theory.

Richard Nelson

Almost at the same time, Nelson presented studies of the US-system (Nelson, 1987 and Nelson, 1988). The focus of the analysis was upon the combined public and private character of technology and the role of, respectively, private firms, government and universities in the production of new technology. It was shown that different industrial sectors use different methods to appropriate the benefits from their innovations.

The approaches of the two authors differ in two important respects. First, while the focus in Nelson's work is upon the production of knowledge and innovation and upon the innovation-system in the narrow sense, Freeman focuses upon the interaction between the production system and the process of innovation. Second, while Freeman applies a combination of organisation and innovation theory – which organisational forms are most conducive to the development and efficient use of new technology? – Nelson's main theoretical tool is related to law and economics – how well can different institutional set-ups take into account and solve the private/public dilemma of information and technical innovation?

The approach of this book is closest to Freeman's because we focus quite a lot upon organisational matters as related to processes of learning, but we also recognise the importance of institutional factors of the kind brought forward by Nelson's work in this area. If one should point to one specific dimension which characterises our approach, it would be the emphasis put upon interactive learning anchored in the production structure and in the linkage pattern of the system of production.

Michael Porter

The recent book by Michael Porter (1990) may be read as a work on national systems of innovation. Porter points to four different determinants affecting the competitiveness of a national industry: Firm strategy, factor conditions, demand conditions and supporting industries.

Actually, Porter refers to the constellation of determinants as a system (Porter, 1990, 75) and he argues that the level at which this system works most strongly is national (and local) rather than international and global. Our approach is akin to Porter's in some respects but different in others. One might say that the basic elements overlap but that their ordering is different.

The main focus in this book is upon explaining learning and innovation and this corresponds to the creation of qualitatively new 'factor conditions'. We regard the economic structure (including 'demand conditions' and 'supporting industries') as one important determinant affecting these processes.

The second fundamental determinant of processes of learning is the institutional set-up and this includes ‘firm strategy’ – including modes of cooperation as well as competition.

The most important difference between our approach and Porter’s may be the level of analysis. While Porter tends to present national systems as environments to single industries involved in international competition, our focus is upon the working of the national system in its own right.¹¹

1.6.2. Different Methods to Analyse National Systems of Innovation

Most discussions of national systems of innovation have been connected to one single or to a comparison of a few specific country cases. In Nelson (1992) fourteen different country specific case studies are presented. One strength of that approach is that the stories told may reflect the complex historical interplay of social, institutional and cultural factors in shaping current systems (Edquist and Lundvall, 1992). The weakness might be the lack of a common and explicit theoretical basis and the fact that the elements brought into the analysis are idiosyncratic reflecting the special interests of each author.

In this book we do not recount any specific case stories even if we refer to relevant cases in order to illustrate some general points. Instead we have tried to present a theoretical perspective which might be used in case studies and to discuss some of the most important sub-systems in the system of innovation. The price we have to pay for choosing this more general approach is a loss in terms of historical richness, especially when it comes to the social and cultural dimensions.

This is the reason why we believe that, together, this book, Nelson (1992) and Porter (1990) may give a good starting point for future work on national systems of innovation.

1.6.3. Conclusions and Open Ends

We sum up collectively in a chapter by Dalum, Johnson and Lundvall (*chapter 14*). In this final chapter we tackle some open-ended and complex policy issues relating to the theme of the book. First, we discuss in general terms the limits for public policy intervention in an economy characterised by innovation and evolution. Second, we look more specifically at what kind of policy prescriptions may be distilled from the interactive learning approach. Finally, we discuss the need for international institutional learning and the role government may play in this context.

Part I:

TOWARD A NEW APPROACH
TO NATIONAL SYSTEMS OF
INNOVATION

Chapter 2

INSTITUTIONAL LEARNING

Björn Johnson

2.1. Introduction

Learning processes, leading to growth in the stock of knowledge, are basic in the dynamics of a modern economy.¹ It will be argued in this chapter that almost all learning processes are interactive, influenced, regarding their content, rate and direction, by the institutional set up of the economy.

When the economy is pictured more as a process of communication and cumulative causation than as an equilibrium system, i.e. from an institutional rather than a neoclassical point of view, learning can be conceptualised as the source of technical innovation. Innovation is then, too, regarded as a process rather than as discrete events uniquely localised in space and time. It follows, it will be argued, that innovation is shaped by institutions and institutional change. It will be suggested that this process can be analysed in terms of national systems of innovation, reflecting that nations differ in terms of institutional set-ups. Furthermore, it will be argued that the relations between institutions and innovation can change, sometimes fundamentally, over time.

Institutions have a strong impact on technical change. However, partly as a consequence of the technical change they shape, a tension between technology and institutions and a pressure for institutional change is often provoked. At the same time institutions are normally quite rigid and do not change easily. The capability of national economies to cope with this problem, i.e. to learn about, adapt and change their institutional frameworks – to engage in ‘institutional learning’ – is important for the development of their international competitiveness.

2.2. The Heritage from Veblen

The impact of institutions on technical change is not a new topic in economic analysis. The ‘institutional drag’ hypothesis for example, often, incorrectly,

ascribed to Thorstein Veblen, is a classic, century old theme in institutional economics. The basic idea is, that institutional factors through their inertia and rigidity retard the dynamics of technical change. Institutions are regarded as inflexible, and institutional change is supposed to be lagging behind technical change. This creates mis-match problems, which often prevent the full realisation of the productive potential of new technologies.²

This hypothesis has lately been revived in different versions. In the analysis of economic policy in the OECD, at least two versions of the idea that institutions retard growth have appeared in the last decade or so. In a series of publications in the latter half of the 1970's, the 'institutional sclerosis' hypothesis was formulated (Johnson, 1981). The argument went, that due to political processes leading to a growing government sector and increasing trade union powers, rigidities were being built into the economic systems of the industrialised Western world, reducing their responsiveness to price signals. At the same time, however, the need for flexible economic systems was assumed to be more pressing than ever. International interdependence was increasing, and there was a strong need for adaptations to radically changing circumstances, including rising energy prices, increasing exports from NIC-countries and new technological opportunities. A growing conflict between an increasing need for change and a decreasing flexibility was assumed to lie behind poor macro-economic performance.

In a more recent analysis of the productivity growth slow-down during the 1970's and 1980's, it is suggested that there is an 'apparent disconnection of technological progress from growth' (OECD, 1987). It is also suggested that there is a need to 'adapt the social and institutional structures to the emerging new techno-economic system'. Rigid organisational structures in firms, inflexible cooperation patterns within and between firms, and inadequate science – and technology policies are supposed to be responsible for inefficiencies in the utilisation of new information technologies. An institutional-technological mis-match is seen as an important explanation of low growth rates in the OECD-area. Similar arguments can be found in the Sundqvist report (OECD, 1988), where lags in translating potential productivity gains stemming from technological advance, especially information technology, into actual productivity gains are associated with the absence of institutional and organisational adaptation.

These examples indicate that the institutional factor is now increasingly taken into account in analyses of technical change. In a way it is not surprising, that the interest in institutional theory is growing during a period, when information technologies are becoming more and more important and information handling activities constitute a growing part of the economy. As we shall see, institutions influence, in fundamental ways, information flows and

learning. These influences go through many channels and have both stimulating and retarding effects. They differ between nations and change over time, and it is utterly misleading to reduce them to sclerotic phenomena of 'lagging' and 'dragging' in the innovation process.

2.3. How Institutions Learn

In this section it is argued that institutions at a very basic level affect all cognitive processes. It will be discussed how learning, growth of knowledge and innovations are influenced by institutions. It is convenient to start with a discussion of the character and functions of institutions.

2.3.1. The Importance of Institutions

The concept of 'institutions' has been defined in many ways. In the institutionalist tradition in economic theory, however, the common idea has been, that societies are characterised by regularities of behaviour and that these are specific to time and place. Economic behaviour is thus instituted, not because of some universal human characteristics, but rather through a process of enculturation (Mayhew, 1987). Culture, which is a local phenomena, is taken as a core concept and its role in shaping human cognitions and actions is emphasised. This was described by Commons (1931) as 'collective control of individual action' and by Veblen (1919) as 'habits of use and wont', and 'habits of thought'.

The simplest form of behavioural regularities are habits. In order to deal with the complexity of everyday life, habits are formed. They provide us with a means of retaining a pattern of behaviour without engaging in global rational calculations involving a vast amount of complex information (Hodgson, 1988). Habits are important in economic analysis, because they relate to a large set of routinised behaviour in the economy. The importance of routines in economic processes was emphasised by Veblen, who actually defined institutions as an 'outgrowth of habit' (Veblen, 1919, 241), and is now increasingly recognised in economic theory, in large part thanks to the work of Nelson and Winter (1982).

When habits and routines become general, common to groups of people, they give rise to different kinds of social regularities in behaviour, such as norms, customs, traditions, rules and laws. These may be formal and explicit as, for example, many laws and administrative guidances, but often they are informal and implicit like common law, everyday customs and moral and social norms. The common feature is that they regulate how individuals and collective groups relate to each other.